**Adventure Works | End-to-End Sales & Business Performance Analytics**

**Power BI · DAX · Data Modeling · Business Intelligence · Sales Analytics**

**Project Overview**

This project uses the Adventure Works sample dataset to develop a multi-page Power BI reporting solution for analysing sales performance, product contribution, profitability, and reseller channel performance.

The objective is to transform transactional sales data into actionable business insights through KPI reporting, time intelligence, comparative analysis, and interactive visualizations.

The report is designed to help business stakeholders understand what is driving revenue, where profitability is strongest, which products require attention, and how reseller performance varies across markets.

**Business Objectives**

- Monitor overall sales, order volume, profit margin, and year-over-year growth.
- Compare current performance against historical benchmarks and sales targets.
- Identify top-performing and underperforming products.
- Analyse profitability by region and product category.
- Evaluate reseller performance across business types and geographies.
- Explore potential drivers of changes in sales and profit.
- Enable interactive exploration through slicers, drill-through, and cross-filtering.

**Dashboard Pages**

**1. Executive Sales Overview**

Provides a consolidated view of business performance through KPI cards, a sales target gauge, monthly trends, category contribution, geographic distribution, sales channels, and a decomposition tree.

**Business questions addressed:**

- How are sales performing?
- How does actual sales performance compare with the selected target?
- Which categories, countries, and products contribute most to revenue?
- How is sales performance distributed across channels?

**2. Product Performance Analysis**

Focuses on product-level performance using Top 10 and Bottom 10 product comparisons, year-over-year sales changes, and monthly sales analysis.

**Business questions addressed:**

- Which products generate the highest sales?
- Which products have low sales or declining performance?
- How has product performance changed compared with the previous year?
- Which products warrant further investigation?

**3. Profitability Analysis**

Examines profit, profit margin, previous-year profit, and profit growth across regions and product categories. The Key Influencers visual supports exploratory analysis of factors associated with changes in profit.

**Business questions addressed:**

- Which regions contribute most to profit?
- Where are profit margins relatively strong or weak?
- How has profitability changed year over year?
- Which attributes are associated with lower profit in the selected data?

**4. Reseller Performance Analysis**

Evaluates the reseller sales channel using reseller sales, business type, monthly trends, geographic distribution, and comparative performance measures.

**Business questions addressed:**

- Which resellers contribute most to sales?
- How does performance vary by reseller business type?
- Which countries and regions contribute to reseller revenue?
- How has reseller sales performance changed over time?

**Tools & Technologies**

- **Power BI Desktop:** Report development and interactive data visualization
- **DAX:** Measures, KPI calculations, time intelligence, and comparative analysis
- **Power Query:** Data preparation and transformation

- **Data modelling:** Relationships between sales, product, date, geography, and reseller entities
- **Business intelligence:** KPI design, performance monitoring, and analytical storytelling

**Key Analytical Measures**

The report incorporates measures for:

- Total Sales
- Total Orders
- Total Profit
- Profit Margin %
- Sales Growth %
- Previous-Year Sales
- Monthly Sales Change %
- Reseller Sales
- Reseller Sales Growth %
- Sales Target Achievement %

Time-based calculations use historical comparisons to support performance analysis. Measures are designed to respond to report filters and slicer selections where appropriate.

**Data Modelling & Methodology**

The solution uses a relational model connecting transactional sales data with descriptive dimensions such as products, dates, geography, and resellers.

The analysis follows these principles:

1. Use measures for aggregations and business KPIs.
2. Apply date-based calculations for month-over-month and year-over-year comparisons.
3. Use dimension attributes to slice and compare performance.
4. Apply consistent filtering and aggregation logic across visuals.
5. Validate totals and calculations before drawing business conclusions.
6. Distinguish missing data from genuine zero values or declines.

**Business Value**

This project demonstrates how Power BI can support:

- Executive performance monitoring
- Product portfolio analysis
- Profitability and regional performance reviews
- Reseller channel evaluation
- Identification of areas requiring further investigation
- Interactive business reporting and decision support

The report is intended as an analytical decision-support tool. Findings from the sample dataset should be validated against business context before being used to make operational decisions.

**Repository Structure**

adventure-works-business-intelligence/

- README.md
- AdventureWorks_Analytics.pbix
- Data Model and Measures.pdf
- Executive Summary.png
- Product Analysis.png
- Profit Analysis.png
- Reseller Analysis.png

**How to Explore the Project**

1. Review the dashboard screenshots in the Screenshots folder.
2. Download the Power BI report file from the PowerBI folder.
3. Open the .pbix file using Power BI Desktop.
4. Explore the report pages, slicers, interactive visuals, and drill-through functionality.
5. Review the measures and data model to understand the calculation logic.

The ability to refresh the report depends on the included data sources and their configuration.

**Validation & Limitations**

The Adventure Works dataset is a sample dataset and does not represent the current performance of a real organization.

Before interpreting the results, validate sales totals, order counts, time-intelligence measures, target definitions, and filter behaviour. Comparisons involving missing prior-period data should be treated carefully.

Sales targets used for demonstration should be clearly distinguished from actual business targets.

**About This Project**

This portfolio project demonstrates practical application of Power BI, DAX, data modelling, and business analysis to a multi-dimensional sales performance problem.

**Focus:** Turning business data into reliable metrics, understandable performance trends, and actionable analytical questions.